# Extreme Value Analysis (EVA) Platform — Complete Implementation Blueprint

## 1. Project Overview

### Objective

Build a production-grade Extreme Value Analysis (EVA) platform for:

- Corrosion prediction
- Wall-loss forecasting
- Risk analysis
- Inspection planning
- Remaining life estimation
- Reliability engineering
- Extreme event prediction

The system should:

- Accept inspection and wall-loss datasets
- Fit statistical extreme value distributions
- Estimate return levels
- Generate confidence intervals
- Validate goodness-of-fit
- Produce engineering reports and visualizations
- Support multi-tenant SaaS deployment

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# 2. Recommended Tech Stack

| Layer | Technology |
|---|---|
| Frontend | Next.js + TypeScript |
| UI | Tailwind + Shadcn UI |
| Backend API | NestJS / Express + TS |
| Statistical Engine | Python FastAPI |
| Statistical Libraries | NumPy, SciPy, Pandas |
| Database | PostgreSQL |
| Queue | BullMQ + Redis |

| Layer | Technology |
|---|---|
| Storage | AWS S3 |
| Deployment | AWS ECS |
| Charts | Plotly |
| Monitoring | Grafana + Sentry |

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# 3. System Architecture

```
Frontend (Next.js)
        ↓
API Gateway (NestJS)
        ↓
Analysis Queue (BullMQ)
        ↓
Python EVA Engine (FastAPI)
        ↓
PostgreSQL + S3
```

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# 4. Complete Application Flow

## Step 1 — User Authentication

### Features

- Login
- Registration
- Role-based access control
- Tenant isolation
- JWT authentication

### Roles

| Role | Permissions |
|---|---|
| Admin | Full access |
| Engineer | Run EVA analysis |
| Viewer | Read-only |

# 5. Main Modules

| Module | Purpose |
|---|---|
| Authentication | Login & security |
| Asset Management | Manage equipment/assets |
| Dataset Upload | Upload inspection data |
| EVA Analysis | Statistical calculations |
| Visualization | Probability & return plots |
| Reporting | PDF/export generation |
| Notifications | Risk alerts |
| Dashboard | Monitoring & KPIs |

# 6. Frontend Page Structure

# 6.1 Login Page

## Purpose

Authenticate users.

## Inputs

| Field | Type |
|---|---|
| Email | Text |
| Password | Password |

## Outputs

- JWT token
- Session creation

# 6.2 Dashboard Page

## Purpose

Display summary statistics.

## Components

- Total assets
- Total inspections
- Risk overview
- Latest EVA runs
- High-risk alerts
- Return-level summaries

## Charts

- Risk trend chart
- Failure probability chart
- Asset condition chart

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# 6.3 Asset Management Page

## Purpose

Create and manage industrial assets.

## Inputs

| Field | Type |
|---|---|
| Asset Name | Text |
| Asset Type | Dropdown |
| Material | Dropdown |
| Location | Text |
| Installation Date | Date |
| Design Thickness | Number |
| Corrosion Allowance | Number |

## Outputs

- Asset record
- Asset metadata

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# 6.4 Dataset Upload Page

## Purpose

Upload wall-loss and inspection datasets.

## Supported Formats

- CSV
- Excel (.xlsx)

## Inputs

| Field | Type |
|---|---|
| Asset Selection | Dropdown |
| Upload File | File |
| Measurement Unit | Dropdown |
| Date Format | Dropdown |

## Required Dataset Columns

| Column | Description |
|---|---|
| inspection_date | Date of inspection |
| wall_loss | Measured wall loss |
| thickness | Remaining thickness |
| location | Measurement location |
| sensor_id | Optional sensor identifier |

## Validation Rules

- No negative wall-loss
- No missing dates

- Numeric validation
- Duplicate check
- Unit consistency

### Outputs

- Dataset ID
- Validation status
- Parsed records

---

## 6.5 EVA Configuration Page

### Purpose

Configure EVA analysis.

### Inputs

| Field | Type |
|---|---|
| Distribution Type | Dropdown |
| Estimation Method | Dropdown |
| Confidence Level | Dropdown |
| Return Periods | Multi-select |
| Peak Extraction Method | Dropdown |
| Outlier Handling | Toggle |

### Distribution Options

- Gumbel
- GEV
- Weibull
- Generalized Pareto (future)

### Estimation Methods

- Maximum Likelihood Estimation (MLE)
- Method of Moments (MoM)

### Peak Extraction Methods

- Annual maxima
- Peak over threshold (future)

### Outputs

- EVA configuration object

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## 6.6 EVA Results Page

### Purpose

Display statistical outputs.

### Sections

1. Distribution parameters
2. Return levels
3. Goodness-of-fit
4. Confidence intervals
5. Probability plots
6. Diagnostics
7. Engineering recommendations

---

# 7. EVA Analysis Engine

## 7.1 Input Data Structure

### Required Input JSON

```
{
  "datasetId": "123",
  "distribution": "gumbel",
  "method": "mle",
  "confidenceLevel": 0.95,
  "returnPeriods": [2, 5, 10, 25, 50, 100]
}
```

---

# 7.2 Preprocessing Stage

## Operations

1. Remove invalid rows
2. Remove NaN values
3. Sort descending
4. Extract maxima
5. Normalize units
6. Validate sample size

## Formula

Number of observations:

$$n = \text{length of dataset}$$

---

# 7.3 Statistical Formulas

# Gumbel Distribution PDF

$$f(x;\mu,\beta) = \frac{1}{\beta}\exp\left(-\frac{x-\mu}{\beta}\right)\exp\left(-e^{-(x-\mu)/\beta}\right)$$

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# Gumbel Distribution CDF

$$F(x) = \exp\left(-e^{-(x-\mu)/\beta}\right)$$

---

# Method of Moments Initialization

## Scale Parameter

$$\beta = \sigma\sqrt{6}/\pi$$

## Location Parameter

$$\mu = \bar{x} - \gamma\beta$$

Where:

| Symbol | Meaning |
|---|---|
| $\sigma$ | Standard deviation |
| $\bar{x}$ | Mean |
| $\gamma$ | Euler constant |

---

# Maximum Likelihood Estimation (MLE)

## Negative Log Likelihood

$$L(\mu, \beta) = n \ln(\beta) + \sum z_i + \sum e^{-z_i}$$

Where:

$$z_i = \frac{x_i - \mu}{\beta}$$

## Optimization Method

Recommended:

- L-BFGS-B
- Nelder-Mead

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# Return Level Formula

Reduced variate:

$$y_T = -\ln\left(-\ln\left(1 - \frac{1}{T}\right)\right)$$

Return level:

$$x_T = \mu + \beta y_T$$

Where:

| Symbol | Meaning |
|---|---|
| T | Return period |
| x_T | Return level |

---

# Probability Plotting Position

$$P_i = \frac{i}{n+1}$$

Reduced variate:

$$y_i = -\ln(-\ln(P_i))$$

---

# Anderson-Darling Test

## AD Statistic

$$A^2 = -n - \frac{1}{n}\sum_{i=1}^{n}(2i-1)\left[\ln(F(x_i)) + \ln(1-F(x_{n+1-i}))\right]$$

## Purpose

Measure goodness-of-fit.

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# Confidence Intervals

## Methods

1. Hessian matrix
2. Fisher Information Matrix
3. Bootstrap resampling

## Recommended Production Method

Bootstrap.

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# 8. EVA Processing Pipeline

## Step-by-Step Flow

```
Upload dataset
    ↓
Validate data
    ↓
Extract maxima
    ↓
Initialize parameters (MoM)
    ↓
Run MLE optimization
    ↓
Compute return levels
    ↓
Compute confidence intervals
    ↓
Run AD test
    ↓
Generate plots
    ↓
Store results
    ↓
Generate report
```

---

# 9. Backend API Design

# Upload Dataset

## Endpoint

```
POST /datasets/upload
```

## Request

Multipart file upload.

### Response

```
{
  "datasetId": "abc123",
  "status": "validated"
}
```

---

# Run EVA Analysis

### Endpoint

```
POST /eva/run
```

### Request

```
{
  "datasetId": "abc123",
  "distribution": "gumbel",
  "method": "mle"
}
```

### Response

```
{
  "runId": "eva_001",
  "status": "processing"
}
```

---

# Fetch EVA Results

### Endpoint

```
GET /eva/:id/results
```

---

# 10. Database Design

## users

| Column | Type |
|---|---|
| id | UUID |
| tenant_id | UUID |
| email | Text |
| role | Text |

---

## assets

| Column | Type |
|---|---|
| id | UUID |
| tenant_id | UUID |
| name | Text |
| material | Text |
| location | Text |
| design_thickness | Float |

---

## inspections

| Column | Type |
|---|---|
| id | UUID |
| asset_id | UUID |
| inspection_date | Date |
| wall_loss | Float |
| thickness | Float |

---

## eva_runs

| Column | Type |
|---|---|
| id | UUID |
| asset_id | UUID |
| distribution | Text |
| mu | Float |
| beta | Float |
| ad_score | Float |
| created_at | Timestamp |

---

## return_levels

| Column | Type |
|---|---|
| id | UUID |
| eva_run_id | UUID |
| return_period | Integer |
| predicted_value | Float |
| ci_lower | Float |
| ci_upper | Float |

---

# 11. Required Visualizations

# Probability Plot

### Purpose

Compare observed vs theoretical values.

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# Return Level Plot

## X-axis

Return period.

## Y-axis

Extreme value.

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# Confidence Interval Plot

## Purpose

Display uncertainty bands.

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# Residual Plot

## Purpose

Check model quality.

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# 12. Report Generation

# Report Sections

1. Asset details
2. Dataset summary
3. Distribution used
4. Estimated parameters
5. Return levels
6. Confidence intervals
7. AD test results
8. Diagnostic plots
9. Engineering interpretation
10. Inspection recommendation

### Export Formats

- PDF
- CSV
- Excel

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# 13. Queue Architecture

### Why Needed

EVA calculations are CPU intensive.

### Flow

```
Frontend
    ↓
API Gateway
    ↓
BullMQ Queue
    ↓
Python Worker
    ↓
Store results
```

---

# 14. Python Statistical Engine Structure

```
eva-engine/
 ├── api/
 ├── services/
 ├── statistics/
 ├── optimization/
 ├── diagnostics/
 ├── plotting/
 ├── bootstrap/
 ├── reports/
 └── utils/
```

---

# 15. Core Python Modules

# preprocessing.py

Responsibilities:

- data cleaning
- sorting
- validation
- maxima extraction

---

# distributions.py

Responsibilities:

- Gumbel PDF
- Gumbel CDF
- GEV support

---

# mle.py

Responsibilities:

- negative log likelihood
- parameter optimization
- convergence checking

---

# diagnostics.py

Responsibilities:

- AD test
- KS test
- residual analysis

---

# bootstrap.py

Responsibilities:

- confidence intervals
- uncertainty estimation

---

# plotting.py

Responsibilities:

- return plots
- QQ plots
- PP plots
- CI visualization

---

# 16. Numerical Stability Requirements

## Critical Safeguards

Use:

```
np.clip()
log-sum-exp
parameter bounds
small epsilon constants
```

## Prevent

- overflow
- underflow
- NaN errors
- optimizer divergence

---

# 17. Production Recommendations

## Mandatory

- Use Python for statistics
- Use PostgreSQL
- Use asynchronous processing
- Add bootstrap confidence intervals
- Add convergence diagnostics
- Add logging
- Add audit trails

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## Avoid

- Running heavy math in frontend
- Using Excel as production engine
- Using raw JavaScript math for MLE
- Ignoring uncertainty estimation

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# 18. Future Enhancements

## Statistical

- Full GEV support
- Bayesian EVA
- POT analysis
- Automatic distribution selection
- Time-dependent EVA

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## AI Features

- Risk prediction AI
- Inspection recommendation AI
- Failure forecasting
- Automated anomaly detection

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## Enterprise Features

- Multi-tenancy
- Team collaboration
- API access
- Audit logging
- Asset hierarchy

---

# 19. MVP Development Order

## Phase 1

Build:

- Authentication
- Dataset upload
- Basic Gumbel EVA
- Return levels
- Probability plots

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## Phase 2

Add:

- Bootstrap CI
- AD test
- Reporting engine
- Dashboard analytics

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## Phase 3

Add:

- Multi-tenancy
- RBAC
- Notifications
- Asset management

---

# Phase 4

Add:

- GEV
- Bayesian analysis
- AI forecasting
- Advanced diagnostics

---

# 20. Final Recommended Architecture

```
Next.js Frontend
        ↓
NestJS API Gateway
        ↓
Redis Queue
        ↓
Python FastAPI EVA Engine
        ↓
PostgreSQL + S3
```

---

# 21. Final Goal

The final system should:

- Accept industrial inspection datasets
- Perform statistically valid EVA analysis
- Estimate extreme future wall loss
- Quantify uncertainty
- Recommend inspection intervals
- Generate engineering-grade reports
- Scale as a multi-tenant SaaS platform
- Support enterprise reliability engineering workflows